

Ivanna Aleman-Coronado

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EDUCATION

The University of Texas at Austin – Austin, TX | Robotics Honors Program

May 2029

B.S. Computer Science, Minor in Robotics

- Relevant Coursework: Computer Architecture, Autonomous Driving, Data Structures, Discrete Math, Linear Algebra, Probability & Statistics

TECHNICAL SKILLS

Programming Languages: Java, Python, C, JavaScript, HTML/CSS, Assembly

Robotics & Systems: ROS Navigation Stack, Boston Dynamics Spot SDK, Foxglove, Unity, Linux, librosa

ML/AI: Reinforcement Learning, Imitation Learning, LLM-based evaluation, K-means/GMM clustering, NumPy/Pandas

Tools & Platforms: Git, VS Code, PyCharm, React, TACC, Figma

Spoken Languages: English (Fluent), Spanish (Fluent)

EXPERIENCE

Undergraduate Research Fellow, Autonomous Mobile Robotics Laboratory (AMRL)

Jan 2026 - Present

Austin, Texas

- Implemented a ROS 2 system on a Boston Dynamics Spot robot to autonomously perform a fully synchronized 50-second dance routine, including a real-time audio beat-detection module and reusable custom move library.
- Resolved low-level body-pose control failures to restore reliable timed motion-command execution.
- Demo adopted as a recurring showcase in lab presentations; currently working on choreography pipeline for outdoor deployment..
- Built a Unity-based co-adaptive driving simulation with PPO training pipelines via PufferLib, distributed across TACC HPC clusters, and an LLM behavior-scoring module feeding a multi-agent decision framework.
- Traced and fixed a ROS message-type mismatch in the lab's person-following stack with a targeted C++ patch to people tracking program, restoring real-time tracking accuracy.

Machine Learning Research Intern, University of Texas at Dallas (StraLiNG Lab)

May 2024 - August 2024

Dallas, Texas

- Built K-means, Gaussian Mixture Model, and One-Class classifier pipelines to detect real-world denial-of-service attacks from network traffic.
- Benchmarked anomaly-based vs. signature-based detection across accuracy, recall, and precision, outperforming traditional signature-based baselines on novel attack patterns.
- Presented findings on ML-based network security to UTD faculty.

PROJECTS

Perception-Driven Social Behaviors on Spot for Public Human-Robot Events

ROS 2, Python, Boston Dynamics Spot SDK

- Engineered an end-to-end ROS 2 pipeline that lets a Spot quadruped perform a hardcoded, fully autonomous 50-second choreography synced to music via a custom beat-detection algorithm.
- Built a modular move library and choreography state machine (`dance_node.py` / `dance_moves.py`), and debugged `spot_ros2` body-pose driver issues to achieve smooth, precisely-timed motion.
- Extending the system into a switchable interactive demo mode: person-following, speech output, and voice-command recognition ("stay", "follow me", "dance") so Spot can respond to untrained users in live settings.
- Targeting outdoor deployment at a public robotics parade and submission to the CoRL demo track

Co-Adaptive Value Alignment for Autonomous Vehicles

Python, Unity, PPO, PufferLib, TACC HPC, Qwen2.5-14B

- Designed a multi-agent reinforcement learning environment in Unity simulating coadaptive merging behavior between human and autonomous drivers, training PPO policies at scale on TACC HPC clusters via PufferLib.
- Integrated a Qwen2.5-14B LLM as a behavior-scoring critic to evaluate driving safety/courtesy in dynamic race-track scenarios, feeding reward signals back into the multi-agent decision-making framework.

LEADERSHIP & AWARDS

- **Scholar, Hispanic Scholarship Fund (HSF)** — Recognized for academic achievement and leadership.
- **Social Officer, The Hispanic Association of Computer Scientists** — Plan and execute social events for 50+ members to build community among Hispanic students in CS and foster peer support and retention in the major
- **Student Ambassador, Texas Exes** — represents the UT Austin alumni association on campus, connecting students to the Texas Exes network (HookedIn, alumni mentorship) and promoting membership benefits and resources to strengthen student-alumni relationships.